

MEMORY MANAGEMENT

1. MAIN MEMORY

Concept

Main Memory (Primary Memory / RAM) is directly accessible by the CPU. It stores:

- Operating System
- Currently executing programs
- Active data

Characteristics

- Volatile
- Faster than secondary storage
- Limited size

Logical vs Physical Address

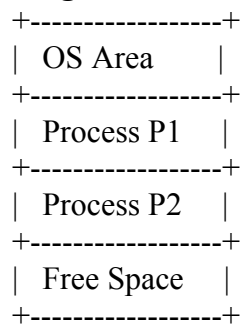
Logical Address: Generated by CPU

Physical Address: Actual location in memory

Address Binding

- Compile time
- Load time
- Execution time

Diagram – Memory View



Numerical Problem 1

Problem:

A system has 1 GB main memory. OS occupies 256 MB. How much memory is available for user processes?

Solution:

Total Memory = 1024 MB

OS Memory = 256 MB

Available = $1024 - 256 = 768$ MB

2. SWAPPING

Concept

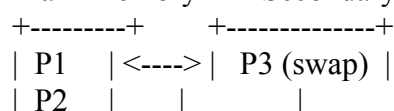
Swapping is a memory management technique where a process is temporarily moved from main memory to secondary storage (swap space) and brought back when required.

Purpose

- Increase degree of multi-programming
- Better memory utilization

Diagram – Swapping

Main Memory Secondary Storage



+-----+ +-----+

Numerical Problem 2

Problem:

If swapping time to disk is 10 ms and swapping back takes 15 ms, find total swap time.

Solution:

Total Swap Time = $10 + 15 = 25$ ms

3. CONTIGUOUS MEMORY ALLOCATION

Concept

Each process is allocated a single continuous block of memory.

Types

Fixed Partitioning

Variable Partitioning

Problems

Internal Fragmentation

External Fragmentation

Diagram – Contiguous Allocation

| OS | P1 | P2 | P3 | Free |

Numerical Problem 3

Problem:

Memory block = 100 KB, Process requires 70 KB. Find internal fragmentation.

Solution:

Internal Fragmentation = $100 - 70 = 30$ KB

4. PAGING

Concept

Paging divides:

Logical memory $\rightarrow$ Pages

Physical memory $\rightarrow$ Frames

Pages are mapped to frames using a **Page Table**.

Advantages

No external fragmentation

Efficient memory utilization

Diagram – Paging

Logical Pages Physical Frames

P0 $\rightarrow$ F3

P1 $\rightarrow$ F1

P2 $\rightarrow$ F0

P3 $\rightarrow$ F2

Address Structure

| Page No | Offset |

Numerical Problem 4

Problem:

Page size = 1 KB, Logical address = 2500 bytes. Find page number and offset.

Solution:

Page Number = $2500 / 1024 = 2$

Offset = $2500 \bmod 1024 = 452$

5. STRUCTURE OF PAGE TABLE

Page Table Entry Contains

Frame number

Valid/Invalid bit
Protection bits
Reference bit

Diagram – Page Table

Page | Frame | Valid

0	3	1
1	1	1
2	0	1
3	-	0

6. SEGMENTATION

Concept

Segmentation divides a program into logical units called segments:

Code, Data, Stack

Each segment has variable size.

Address Format

| Segment No | Offset |

Diagram – Segmentation

Segment Table

+-----+	+-----+	+-----+
Seg #	Base	Limit
+-----+	+-----+	+-----+

Numerical Problem 5

Problem:

Segment base = 4000, offset = 300. Find physical address.

Solution:

Physical Address = $4000 + 300 = 4300$

7. SEGMENTATION WITH PAGING

Concept

Logical address → Segment → Page → Frame

Combines benefits of both

Diagram – Segmentation with Paging

Logical Address

Segment → Page → Frame → Physical Address

8. VIRTUAL MEMORY

Concept

Virtual memory allows execution of programs larger than physical memory by using disk as extension of RAM.

Advantages

Large address space

Better CPU utilization

9. DEMAND PAGING

Concept

Pages are loaded into memory **only when required**.

Page Fault

Occurs when required page is not in memory.

Diagram – Demand Paging

CPU → Page Table → Page Fault → Disk → Memory

Numerical Problem 6

Problem:

Page fault service time = 20 ms, memory access = 1 ms. Effective access time?

Solution:

EAT $\approx 20 + 1 = 21$ ms

10. COPY ON WRITE (COW)**Concept**

Used during process creation (fork)

Pages are shared until modification

Advantage

Saves memory

Faster process creation

Diagram – Copy on Write

Parent Page <--- Shared ---> Child Page

(Write causes copy)

11. PAGE REPLACEMENT**Concept**

When memory is full and a page fault occurs, one page must be replaced.

Algorithms

FIFO

LRU

Optimal

Numerical Problem 7 (FIFO)**Problem:**

Page reference string: 7, 0, 1, 2, 0, 3

Frames = 3

Solution:

FIFO Page Faults = 5

12. ALLOCATION OF FRAMES**Types**

1. Equal Allocation

2. Proportional Allocation

Numerical Problem 8**Problem:**

Total frames = 100, Two processes of sizes 10 KB and 30 KB.

Solution:

$P1 = (10 / 40) \times 100 = 25$ frames

$P2 = (30 / 40) \times 100 = 75$ frames

13. THRASHING**Concept**

Thrashing occurs when system spends more time paging than executing processes.

Cause

Too many processes,

Insufficient frames

Diagram – Thrashing

High Page Fault Rate

Low CPU Utilization

Prevention

Working set model

Load control

END OF NOTES